

Lecture 11-20

determinism / classical

$$\mathcal{L}^t(x, y) = \delta(y - f^t(x))$$

evolution operator

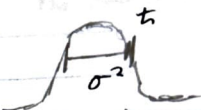
→ linear, a matrix

→ applies to vector coords

→ admits group theoretic properties

could be a dist:

(QM, stochastic)



→ But what do we want? → Predictions, observables

→ we have $\bar{a}$: averages over $\underline{a}$ (expectation values)

space

$$\int dx p(x) a(x) = \langle a \rangle_p$$

↳ look anywhere in state space, at the natural measure $p(x) dx$

time

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t d\tau a(f^\tau(x_0)) = \bar{a}_{x_0}$$

space-time

$$\langle a \rangle = \lim_{t \rightarrow \infty} \int dx \frac{1}{t} \int_0^t d\tau a(f^\tau(x_0))$$

Chaos Book ch. 20 Averaging

Birkhoff: consider an observable $a(x(\tau))$

Flow (continuous t) $A(x_0, t) = \int_0^t d\tau a(x(\tau))$

(average over space times the time interval)

Map (discrete t) $A(x_0, n) = \sum_{m=0}^n a(x_m)$

Consider the periodic orbit p

$$A_p = \int_0^T dx a(x(z))$$

$x(t) \in M_p$ ^{set describing orbit on the manifold}

$$\sum_{M=0}^{M_p} a(x_M)$$

$$a_p = \frac{1}{T} A_p$$

$$= \frac{1}{n_p} A_p$$

intrinsic to the system; not dependent on coordinates

→ "I think it was invented by Laplace in 18-something"

inverse temperature

$$\langle e^{B \cdot A} \rangle$$

can be vectors

↔ stat mech formalism + Gibbs
even though our system is deterministic: this is because the treatment is good at summing many small contributions (probabilities)

$$\langle e^{B \cdot A} \rangle_t = \frac{1}{|M|} \int dx e^{B \cdot A(x,t)}$$

discretize

$$\langle A \rangle_t = \frac{\partial}{\partial B} \langle e^{B \cdot A} \rangle \Big|_{B=0}$$

$$\langle e^{B \cdot A} \rangle_t \approx e^{t S(B)}$$

something (sum of exponentials becomes exp carrying the dominant behavior)

if you have a 'healthy' observable (like temperature), then the quantities above are bounded:

$$t_{amin} \leq A^t \leq t_{amax}$$

→ out of this, comes an expression (Helmholtz theorem)

$$\langle e^{\beta \cdot A^t} \rangle \simeq k e^{t S(\beta)}$$

take log

$$\log \langle e^{\beta \cdot A^t} \rangle \simeq t S(\beta) \cdot \text{something}$$

$$\Rightarrow \lim_{t \rightarrow \infty} \frac{1}{t} \log \langle e^{\beta \cdot A^t} \rangle = S(\beta)$$

$$\Rightarrow S(\beta) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \langle e^{\beta \cdot A^t} \rangle$$

→ so why is this good?

↳ we know if we differentiate wrt to one of our observable components we get

$$\frac{\partial S(\beta)}{\partial \beta_i} = \lim_{t \rightarrow \infty} \frac{1}{t} \langle A^t \rangle$$

↓

$t a$

$$= \lim_{t \rightarrow \infty} \frac{1}{t} \cdot (at) = \underline{\langle a \rangle_i!}$$

→ Hence, we can directly relate these abstract averages over state space to the observables themselves

→ 1st derivative gives you the expectation, higher derivatives give you the covariance b/t different observables

→ So how do we compute these? Return to the evolution operator:

Ruelle → $\mathcal{L}^t(x, y)$ → e-values $s_0, s_1, \dots$
(Perron-Frob) "matrix" $\Rightarrow \{e^{s_0 t}, e^{s_1 t}, \dots\}$

Perron-Frobenius Theorem: if all elements of the matrix are positive/finite, then the leading eigenvalue $s_0 > s_1$ is strictly greater

→ but we know that $s_0 = 0$. This implies (why?)

$\Rightarrow s_1 < 0$, and we can compute the spectrum of $\mathcal{L}^t$!

→ in general, the e-values will be functions of the system parameters

So we are talking about the eigenvalues

$$\lambda_0 = e^{s_0}, \lambda_1 = e^{s_1}, \dots$$
$$\lambda_n(t) = e^{s_n(t)}$$

So, since along the invariant measure $\tilde{\nu}_1$, the system does not contract, we know

$$\lambda_0 = 1 \Rightarrow 1 = e^{s_0} \Rightarrow s_0 = 0$$

→ Then, along all other eigendirections, the eigenfunctions (values) will contract due to dissipation

Spectrum:

$$L^t(x, y) : \left\{ \boxed{S_0}, S_1, S_2, \dots \right.$$

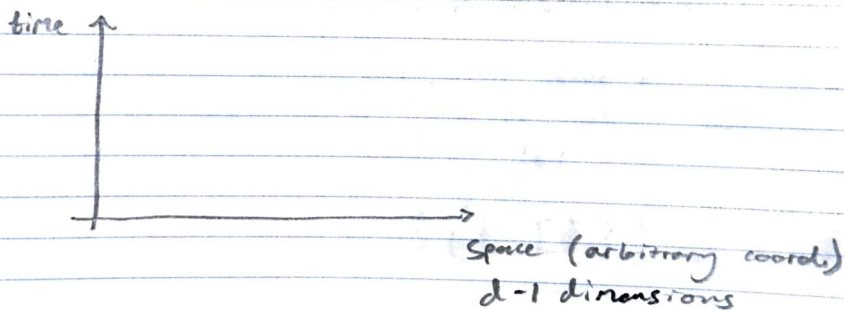
invariant measure
 $S_0 = 0$

collapse as $t \rightarrow \infty$ due to dissipation

Perron-Frobenius $\Rightarrow S_0 = 0$, so we have that $S_1 < 0$, as well as $S_2, \dots, S_n$

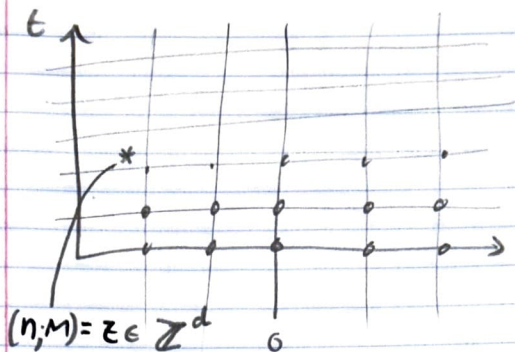
Deterministic Field Theory

The Law: $F = 0$ ← determinism



→ properties of the law at each point in space create a field; think line blades of grass flowing in a meadow, constituting a field

→ Now, say we want to "know" the system, so we discretize



hyper-cubic lattice
array upon which
we define our
fields

space (d-1 dims)

$$(n, M) = z \in \mathbb{Z}^d$$

$$F[\Phi]_z = 0$$

$$\begin{array}{ccccc} \phi_{i-1} & \phi_{i,0} & \phi_{i,1} & & \\ \phi_{0-2} & \phi_{0-1} & \phi_{0,0} & \phi_{0,1} & \phi_{0,2} \\ \phi_{-1} & \phi_{-1,0} & \phi_{-1,1} & & \end{array}$$

$$\phi = \{\phi_z\}$$

→ How do we respect the law above using our field?

$$\delta(F_{0,0}[\Phi]) \cdot \delta(F_{0,1}[\Phi]) \cdot \dots$$

on every lattice site, we have an infinitely-sharp Dirac-delta function which satisfies the condition exactly. We let

$$\delta(F[\Phi]) = \delta(F_{0,0}[\Phi]) \cdot \delta(F_{0,1}[\Phi]) \cdot \dots$$

→ can be considered a special case of statistical mechanics